Apply filters to SQL queries

Project description

In this activity, I serve as a security professional at a large organization. My job is to investigate security issues to help keep the system secure. My task is to examine the organization's data in their `employees` and `log_in_attempts` tables. To do this, I will utilize SQL filters to retrieve records from different datasets and investigate the potential security issues.

Retrieve after hours failed login attempts

An incident occurred after business hours (18:00) and it is my job to investigate it. I will identify this suspicious behavior by using the following commands:

```
SELECT *
FROM log_in_attempts
WHERE login_time > '18:00'
  AND success = failure;
```

The query above selects all from the `log_in_attempts` table. Then, the `WHERE` clause I included filters my results to showcase `login_time` that was after 18:00 and resulted in failure.

Output: 19 failed login attempts occurred after 18:00

Retrieve login attempts on specific dates

Another suspicious event occurred on 2022-05-09. To investigate this event, I will review all login attempts which occurred on this day and the day before. This is showcased by the following:

```
SELECT *
FROM log_in_attempts
WHERE login_date = '2022-05-09'
   OR login_date = '2022-05-08';
```

The query above selects all from the `login_in_attempts` table and filters it using the `WHERE` clause to only include logins made on or before `'2022-05-09'.`

Output: 75 login attempts made

Retrieve login attempts outside of Mexico

There's been suspicious activity with login attempts, but the team has determined that this activity didn't originate in Mexico. Now, I need to investigate login attempts that occurred outside of Mexico. I will use filters in SQL to create a query that identifies all login attempts that occurred outside of Mexico:

```
SELECT *
FROM log_in_attempts
WHERE country NOT LIKE '%MEX%';
```

This selects all from the `login_in_attempts` table and filters it using the `WHERE` clause to only include countries that are not Mexico (`NOT LIKE`).

Output: 144 login attempts made outside of Mexico

Retrieve employees in Marketing

Next, my team wants to perform security updates on specific employee machines in the Marketing department. I am responsible for getting information on these employee machines and will need to query the employees table. I will utilize the following filters in SQL to create a query that identifies all employees in the Marketing department for all offices in the East building:

```
SELECT *
FROM employees
WHERE department LIKE '%Marketing%'
  AND office LIKE 'East%';
```

This query selects all from the `employees` table and filters it using the `WHERE` clause to only include the Marketing department and East office. The `LIKE` operator returns every instance that matches.

Output: Showcases `elarson` as the username for the first employee listed.

Retrieve employees in Finance or Sales

Now I need to perform a different security update on machines for employees in the Sales and Finance departments. I will use the following filters in SQL to create a query that identifies all employees in the Sales or Finance departments:

```
SELECT *
FROM employees
WHERE department LIKE '%Finance%'
   OR department LIKE '%Sales%';
```

This query selects all from the `employees` table and filters it using the `WHERE` clause to only include the Finance and Sales departments. The `LIKE` operator returns every instance that matches.

Output: Showcases `Irodiqu` as the username for the first employee listed.

Retrieve all employees not in IT

Finally, I need to use the following filters in SQL to create a query which identifies all employees not in the IT department:

```
SELECT *
FROM employees
WHERE department NOT LIKE '%Information Technology%';
```

This query selects all from the `employees` table and filters it using the `WHERE` clause to only include departments that are not Information Technology. The `NOT LIKE` operator returns every instance that is not Information Technology.

Output: Showcases 165 employees with `elarson` as the username for the first employee listed.

Summary

This activity allowed me to enhance my SQL querying skills to quickly filter through company data. I utilized operators such as AND, OR, NOT, and LIKE to filter for relevant information I

needed. These queries not only showcased potential data breaches, but also allowed me to utilize SQL to find important data. These are valuable skills that I can utilize as a potential cybersecurity analyst in the industry.